SINKGRAB · FEASIBILITY ASSESSMENT

SinkGrab feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 (SKG-DEC-001 O2, O4, O5) written in: bentonite annulus, crank slip clutch and load indicator, four people at the T-bar
0.1	2026-10-24	Amish Chadha	First feasibility assessment (kit 1.9.1 upgrade)

SinkGrab feasibility assessment

SinkGrab is feasible to build, and two parts of its method that did not work on paper as drawn now carry the fixes Amish chose on 2026-10-25, each still to be shown at TRL 4: the caisson rings sinking 3 m without de-watering (STEM S21, PR-001), where the central unaided friction estimate stops them, now with a bentonite slurry in the overcut annulus (SKG-DEC-001 O2); and the overload device, where no shear pin fitted between the working pull and the tripod's rating (S6, S7, PR-007), now a friction slip clutch on the crank shaft with a load indicator at the dead end, whose 6 % window is shown on CalRig (O4). The mechanism and the lifting gear fit together in the detailed D3 model; the rest can only be settled in a test pit and a partner well. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design: grab	A one-line clamshell with a 3:1 tackle lifts 21 L a bite at an assumed 75 % fill, inside the tripod's rating	Medium	Closing margin over passive drag in loose sand is about 0.9 to 1.8 after the independent review (S2); fill in denser soil and the head-weight opening (S3) need a bite trial	SKG-STM-001 S1 to S5; SKG-CAL-001 A to C
Design: scraper	A pole scraper opened by weight, turned from the surface, folds to pass a 0.8 m ring	Medium	Turning effort of 142 N each for two people exceeded sustained limits (B4); four people now turn it, 71 to 84 N each, since a 2.0 m T-bar does not clear the tripod legs (O5, 2026-10-25); the pivot change of K6 keeps the fold inside the bore (387 mm)	SKG-STM-001 S17 to S20, B4; SKG-CAL-001 H
Design: sinking the rings	Self-weight and eight saddles against skin friction, with a 40 mm overcut lubricated by bentonite slurry (O2, 2026-10-25)	Low	Sinks only below 1.66 kPa of friction; the central unaided estimate (2 kPa) exceeds it and soil in full contact would give about 25 kPa; the lubricated friction is unknown until the test pit	SKG-STM-001 S21; SKG-CAL-001 I
Design: lifting and overload limits	Hand capstan with a 4:1 chain, weighted band brake, crank slip clutch and dead-end load indicator (O4, 2026-10-25)	Medium	Brake geometry corrected (K2); the clutch caps the capstan pull at 1,472 N and carries the steady full-grab pull with a 6 % margin on the third layer on a ±5 % setting, to show on CalRig; proof at 1.25 x the rated load, 188 kg	SKG-STM-001 S4 to S11, E2; SKG-CAL-001 C to G
Materials	Common S235 and S275 steel section and plate, bought bearings, sheaves, chain and polyester rope	High	Wear-resistant plate for lips and toes may need a stockholder in a city; lead-free paint on wetted parts	BOM; SKG-REG-001 G5, G15
Manufacturing and assembly	Stick welding, drilling, plate rolling for the shells, profile cutting for the ratchet, pawl and end plates	Medium	Plate rolls and a profile cutter are not in every village workshop; 2 mm tube welding needs skill; 21 detail fixes are now in the model and drawings	SKG-BLD-001; SKG-DEC-001 K6 to K21; `cad/fidelity.json`
Power and controls	Human power only: two people at the cranks, three during the hoist, two at the T-bar	Medium	Hoist power sets the cycle (PR-004, about 0.2 min over); no engine, so no exhaust in the shaft	SKG-CAL-001 D, J; STEM S8, S9
Supply of parts	Bearings, chain, sheaves, rope, shackles and detectors from general suppliers; HatchSide tripod shared	High	Gas detectors with carbon dioxide may need a city supplier (BOM line 35)	BOM; SKG-MKT-001
Safety case	Nobody in the well; doors over the shaft; overload	Medium	Depends on crews keeping the stop rules; the HatchSide tripod is rated for material handling only	SKG-RSK-001; SKG-BLD-001 section 6

Aspect	Assessment	Rating	Risks and notes	Evidence
	limits; air test with carbon dioxide; proof load			

Ratings: High, Medium, Low.